

Article

Industrial Intelligent Transformation and Youth Employment and Career Development: A Regional Socio-Economic Systems Analysis of Zhejiang, China

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Abstract

The coexistence of severe youth employment pressure and persistent shortages of skilled industrial labor has become a defining dilemma of regional socio-economic systems undergoing rapid industrial upgrading. From a socio-economic systems perspective, industrial intelligent transformation constitutes a technological subsystem shock whose youth-employment consequences are shaped by skill reconfiguration and multi-level employment governance rather than by technology alone. Using panel data on 668 A-share listed firms registered in Zhejiang province—China's pioneer region of large-scale “machine substitution” and digital-economy development—from 2011 to 2023, we examine how firm-level industrial intelligent transformation affects youth employment and youth career development. We employ two-way fixed-effects models, mediation and moderation analyses, and a battery of robustness checks, including propensity score matching and instrumental-variable estimation. We find, first, that industrial intelligent transformation significantly increases both the youth-employment share and the representation of young executives, indicating that task creation and organizational renewal outweigh displacement for young workers. Second, this effect operates through two resource-reconfiguration channels: a skill-restructuring effect through upgrading of the firm skill structure and an expansion effect through productivity-driven growth. Third, multi-level employment governance plays a decisive moderating role: firm training investment, regional public employment services, and digital-economy policy pilots significantly strengthen the positive effect of intelligent transformation on youth employment. Moreover, the effect is stronger for manufacturing, larger, non-state-owned firms and firms located in the digital core cities. These findings portray youth employment as an emergent outcome of a regional socio-economic system and inform coordinated countermeasures for youth-inclusive industrial upgrading.

Keywords: industrial intelligent transformation; youth employment; career development; regional socio-economic systems; skill restructuring; multi-level governance; robot penetration; Zhejiang

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1. Introduction

Youth employment and career development have become one of the most pressing challenges confronting regional socio-economic systems in the digital era. The International Labour Organization estimates that the global youth unemployment rate remains substantially higher than the adult rate, and that a large share of young people are not in employment, education, or training [1–3]. In China, the urban youth unemployment rate reached historically high levels in recent years, while manufacturing regions simultaneously report persistent shortages of skilled industrial labor [4,5]. This coexistence of “employment difficulty” among educated young job seekers and “recruitment difficulty” among upgrading firms constitutes the youth employment dilemma: a structural mismatch between the tasks that regional industrial systems create and the skills, expectations, and career prospects of the young labor force [6,7]. At the same time, the rapid diffusion of industrial robots and intelligent manufacturing is transforming the structure of labor demand, automating routine production tasks while creating new digitally intensive occupations [8,9]. How industrial intelligent transformation reshapes the demand for young workers and their career prospects is therefore a question of first-order importance for regional employment governance and youth-inclusive industrial upgrading [10,11].

The prevailing public debate frames intelligent transformation mainly as a threat to employment, emphasizing the displacement of workers by machines [12]. Yet a growing body of firm-level evidence suggests a more nuanced picture: robot adoption and intelligent manufacturing can complement labor, raise productivity, and generate new roles associated with programming, supervision, data analysis, and digitally enabled services [13,14]. For young workers in particular, the net effect is theoretically ambiguous. On the one hand, youth are overrepresented in entry-level routine occupations that are most exposed to automation [10]. On the other hand, as digital natives with recently acquired human capital, young workers hold a comparative advantage in the new digitally intensive tasks that intelligent transformation reinstates, and technological renewal may open faster promotion tracks for young technical and managerial talent [15,16]. Which of these forces dominates is ultimately an empirical question that depends on how firms reconfigure their skill structures and how multi-level governance mechanisms channel technological change toward inclusive value creation.

This study addresses this question from a socio-economic systems perspective. Socio-technical and socio-economic systems theory conceptualizes the region and the firm as adaptive systems in which technological and social subsystems must be jointly optimized rather than changed in isolation [17–19]. From this perspective, industrial intelligent transformation constitutes a technological subsystem transformation; the employment governance architecture—firm-level training investment, regional public employment services, and digital-economy policy pilots—constitutes the social and institutional subsystem; and youth employment and career development represent system-level labor outcomes that emerge from the interaction of these subsystems [20,21]. Accordingly, the relationship between intelligent transformation and youth employment is not a simple technology-labor substitution, but a socio-economic adaptation

process shaped by skill reconfiguration, productivity-driven expansion, and multi-level governance.

Zhejiang province provides a representative regional laboratory for this inquiry. As one of China's most developed provincial economies and the national pioneer of large-scale “machine substitution” (jiqui huanren) in manufacturing since 2013, Zhejiang has experienced both intensive industrial intelligent transformation and acute youth-employment pressure [22,23]. The province designated the digital economy as its “Number One Project” in 2017, was selected as the national common prosperity demonstration zone in 2021, and has committed to building a “youth-development-oriented province”, making the coordination between industrial upgrading and youth opportunity an explicit policy objective [24,25]. The coexistence of a large private manufacturing sector, rapid robot adoption, a dense cohort of young graduates attracted by the digital economy, and active employment-governance experimentation makes Zhejiang an informative setting for examining how industrial intelligent transformation, skill restructuring, and multi-level governance jointly shape youth employment and career development in a regional socio-economic system.

Existing research has examined the employment effects of robots and automation, the determinants of youth employment, and regional socio-economic system analysis, respectively. However, these streams have largely developed in parallel. The task-based literature on automation and jobs rarely distinguishes age-specific effects or connects them to career advancement within firms [10,13,26]. The youth-employment literature emphasizes education, institutions, and macroeconomic conditions but pays limited attention to firm-level technology adoption and its governance context [1,4]. The regional socio-economic systems literature analyzes urban, regional, and industrial dynamics but seldom traces the youth-labor-market consequences of industrial intelligent transformation [21,27]. Less attention has thus been paid to how intelligent transformation, skill restructuring, productivity-driven expansion, and multi-level employment governance interact within a unified regional system to shape youth employment and career development.

Based on the above analysis, we use panel data on 668 A-share listed firms registered in Zhejiang province from 2011 to 2023 to examine the impact of firm industrial intelligent transformation on youth employment and youth career development, its underlying mechanisms, and the moderating role of multi-level employment governance. The empirical results show that, first, industrial intelligent transformation significantly increases both the youth-employment share and the representation of young executives, and these findings remain robust after a series of robustness checks, including propensity score matching and instrumental-variable estimation. Second, this positive effect is primarily transmitted through two channels: a skill-restructuring effect and an expansion effect. On the one hand, intelligent transformation upgrades the firm skill structure, raising the relative demand for young, high-skill labor. On the other hand, intelligent transformation raises labor productivity and firm growth, expanding total hiring capacity in which young workers are disproportionately represented. Third, multi-level employment governance plays a crucial moderating role: firm training investment, regional public employment services, and digital-economy policy

pilots significantly strengthen the positive effect of intelligent transformation on youth employment. Furthermore, the effect is significantly stronger for manufacturing, larger, and non-state-owned firms and for firms located in Zhejiang's digital core cities.

Our contributions are threefold. First, we clarify the theoretical novelty of framing the youth employment dilemma as an emergent outcome of a regional socio-economic system. Prior studies mainly examine the aggregate or task-level employment effects of automation, or analyze youth unemployment from macroeconomic and institutional perspectives. We extend these studies by showing how a technological subsystem shock, firm-level resource reconfiguration, and multi-level governance jointly determine whether industrial upgrading absorbs or excludes young workers, thereby connecting the automation-employment debate with regional socio-economic system analysis [17,21].

Second, we extend the literature on the labor-market effects of industrial intelligence by identifying two resource-reconfiguration mechanisms through which intelligent transformation promotes youth employment—a skill-restructuring channel operating through the upgrading of the firm skill structure and an expansion channel operating through productivity-driven growth—and by tracing the career-development consequences of intelligent transformation through the representation of young executives. This reveals not only whether firms hire young workers under technological change, but also whether young talent advances within transforming organizations [16,28].

Third, we enrich research on employment governance by showing that firm training investment, regional public employment services, and digital-economy policy pilots strengthen the positive effect of intelligent transformation on youth employment. This highlights multi-level governance as a key condition that helps regional systems translate industrial upgrading into youth-inclusive opportunity, and provides direct evidence for the design of coordinated countermeasures to the youth employment dilemma [29–31].

The rest of this study is organized as follows. Section 2 presents the institutional background, theoretical analysis, and hypotheses. Section 3 describes the research design. Section 4 reports the empirical results. Section 5 discusses the findings and limitations. Section 6 concludes the study and presents the implications.

2. Institutional Background, Theoretical Analysis, and Hypotheses

2.1. The Youth Employment Dilemma and Industrial Intelligent Transformation in Zhejiang

Zhejiang is one of China's most market-oriented provincial economies, characterized by a dense private manufacturing sector, extensive county-level industrial clusters, and an advanced digital economy anchored by the platform ecosystems of Hangzhou [22]. Facing rising labor costs and recruitment difficulties in manufacturing, the provincial government launched a province-wide “machine substitution” program in 2013, subsidizing the large-scale replacement of routine manual tasks with industrial robots and intelligent equipment; Zhejiang has since ranked

among the leading provinces in robot installation in China [14,23]. In 2017, the province designated the digital economy as its “Number One Project”, and in 2021 it was selected as the national common prosperity demonstration zone, which explicitly links industrial upgrading to inclusive employment and income growth [24].

Against this backdrop, the regional youth labor market exhibits a pronounced structural dilemma. On the demand side, upgrading firms report shortages of technicians, engineers, and digitally skilled operators, and the skill requirements of newly created positions rise rapidly with intelligent transformation [6,15]. On the supply side, a growing cohort of young graduates—attracted by Zhejiang's digital economy and quality of life—faces employment difficulty, underemployment, and slow early-career advancement, while showing a declining willingness to enter traditional manufacturing occupations [5,7,32]. The dilemma is thus not a simple deficiency of labor demand, but a system-level mismatch among task creation, skill formation, and career expectations within the regional socio-economic system. Whether industrial intelligent transformation aggravates or alleviates this mismatch—and under which governance conditions—is the central question of this study.

2.2. Industrial Intelligent Transformation and Youth Employment: A Socio-Economic Systems Perspective

Socio-technical and socio-economic systems theory provides an overarching basis for understanding the relationship between industrial intelligent transformation and youth employment. It emphasizes that organizational and regional transformation depends on the alignment and joint optimization of social and technical subsystems rather than on technological change alone [17–19]. In the context of this study, intelligent transformation reconfigures the technical subsystem by embedding programmable automation, adaptive production, and data-driven control into manufacturing and service processes. This technological reconfiguration in turn alters the composition of tasks the firm performs and, therefore, the composition of the workforce it demands. Youth employment can accordingly be understood as a system-level outcome that emerges from the interaction between the technical subsystem (intelligent production) and the social subsystem (skill formation, employment governance, and organizational routines) [20,33].

The task-based framework of technological change clarifies the mechanisms through which intelligent transformation affects labor demand. In this framework, technology exerts two opposing forces: a displacement effect, whereby automation substitutes machines for labor in tasks previously performed by workers, and a reinstatement (or productivity and creation) effect, whereby technology generates new tasks and expands labor demand in activities where labor retains a comparative advantage [8,26]. Cross-country and firm-level evidence indicates that the net employment effect of robot adoption is highly heterogeneous: negative in some aggregate settings [9], but frequently positive at the firm level, where adopters expand output, upgrade products, and hire complementary labor [13,14,34].

For young workers, this task reallocation carries a distinctive age-specific logic. First, the new tasks reinstated by intelligent transformation—such as robot programming and maintenance, industrial data analysis, digital quality

control, and human-machine collaboration—are intensive in recently acquired digital skills, in which young workers hold a comparative advantage [10,15]. Second, human capital theory suggests that younger workers have a longer horizon over which to amortize investments in firm- and technology-specific skills, making them attractive candidates for the newly created technical positions [35]. Third, from a dynamic capabilities perspective, firms undertaking intelligent transformation must sense, seize, and reconfigure resources to exploit new opportunities; recruiting young, digitally fluent talent is a low-cost, high-flexibility way to build the absorptive capacity required for technological renewal [36]. At the same time, intelligent transformation may displace some young workers from entry-level routine positions, and firms facing adjustment costs may delay hiring. Whether the creation force dominates is therefore an empirical question; on balance, in a region whose transformation is growth-oriented rather than purely cost-cutting, we expect the creation and reinstatement forces to dominate for young workers.

H1a. *Industrial intelligent transformation increases the firm's youth-employment share.*

Beyond the quantity of youth employment, intelligent transformation may also reshape the career development of young workers within the firm. Organizational renewal driven by new technology tends to revalue the knowledge base of the workforce: seniority-based advantages erode when production logic shifts toward digital and data-driven competencies, opening promotion opportunities for younger technical and managerial talent [16,28]. Firms undergoing intelligent transformation are more likely to establish new digital business units, smart-manufacturing divisions, and data teams, which create management positions disproportionately filled by young professionals [37]. From a socio-economic systems perspective, the advancement of young managers reflects the adaptation of the organizational authority structure to the requirements of the transformed technical subsystem. We therefore expect intelligent transformation to increase the representation of young executives, an observable marker of youth career development.

H1b. *Industrial intelligent transformation increases the representation of young executives, promoting youth career development.*

2.3. Mediating Mechanisms: Skill Restructuring and Productivity-Driven Expansion

From a socio-economic systems perspective, firms translate technological change into labor outcomes through the reconfiguration of both their human capital and their scale of activity. Accordingly, this study identifies two mediating mechanisms: a skill-restructuring effect through the upgrading of the firm skill structure and an expansion effect through productivity-driven growth.

The first mechanism concerns skill restructuring. Skill-biased technological change theory holds that advanced technologies are complementary to high-skill labor and substitutable for low-skill routine labor, thereby raising the relative demand for skilled workers [15,26]. Intelligent transformation intensifies this bias: it automates codifiable routine tasks while increasing the marginal product of workers who can program, deploy, maintain, and

collaborate with intelligent systems [13,38,39]. As firms upgrade their production systems, they raise the share of technically educated, high-skill employees. Because the most recent cohorts of graduates are the primary carriers of digital and analytical skills, an upgrading of the firm skill structure translates into greater relative demand for young, high-skill workers [6,15]. In socio-technical terms, skill upgrading reflects the adaptation of the human-capital component of the social subsystem to the requirements of the intelligent technical subsystem. We therefore expect intelligent transformation to promote youth employment partly by upgrading the firm skill structure.

H2a (Skill-Restructuring Effect). *Industrial intelligent transformation promotes youth employment by upgrading the firm skill structure.*

The second mechanism concerns productivity-driven expansion. The productivity effect in the task-based framework implies that automation lowers production costs, improves quality and flexibility, and thereby expands product demand and firm scale [8,34]. Firm-level studies find that robot adopters grow faster in output and employment than non-adopters, as scale expansion offsets within-task displacement [13,14]. Expanding firms recruit disproportionately from the external labor market, where young job seekers are concentrated; growth also relaxes the budget constraints that otherwise bias hiring toward experienced incumbents [11,40]. In socio-economic system terms, the expansion effect couples the technical subsystem to the regional youth labor market through the growth of the firm's value-creation activities. We therefore expect intelligent transformation to promote youth employment partly by raising labor productivity and expanding firm scale.

H2b (Expansion Effect). *Industrial intelligent transformation promotes youth employment by raising labor productivity and expanding the firm's scale of activity.*

2.4. The Moderating Role of Multi-Level Employment Governance

From a socio-economic systems perspective, the extent to which intelligent transformation is translated into youth employment depends on the social and institutional subsystem that governs skill formation and labor-market matching. Organizational information processing theory argues that environmental and technological change increases the demand for information-processing and coordination capacity, and that organizations embedded in stronger governance arrangements are better able to absorb and exploit new technologies [33]. We focus on three governance mechanisms that span the firm, regional, and policy levels emphasized by regional socio-economic system analysis: firm training investment, regional public employment services, and digital-economy policy pilots.

Firm training investment captures the organization's internal capability to convert new technology into usable skills. Firms that invest more in employee training can rapidly equip newly hired young workers with the firm- and technology-specific competencies required by intelligent production, lowering the effective cost of youth recruitment and reducing skill-mismatch frictions [35,41]. Training investment also signals a human-capital-augmenting rather than labor-displacing transformation strategy, which encourages complementary hiring. We therefore expect firm training

investment to amplify the positive effect of intelligent transformation on youth employment.

H3a. *The positive effect of industrial intelligent transformation on youth employment is positively moderated by firm training investment.*

Regional public employment services capture the quality of the local labor-market matching infrastructure, including public job-placement platforms, vocational training programs, and employment subsidies [29,30]. Effective public employment services reduce search and matching costs between upgrading firms and young job seekers, provide complementary vocational training that aligns graduate skills with the requirements of intelligent manufacturing, and cushion the adjustment costs of displaced routine workers [42]. In system terms, public employment services strengthen the coupling between the regional skill-formation subsystem and the firm's technical subsystem. We therefore expect stronger regional public employment services to amplify the youth-employment benefits of intelligent transformation.

H3b. *The positive effect of industrial intelligent transformation on youth employment is positively moderated by regional public employment services.*

Digital-economy policy pilots capture the supportiveness of the institutional environment for intelligent transformation, including national big-data comprehensive pilot zones and new-generation artificial intelligence innovation and development pilot zones [25,31]. Such pilots provide regulatory clarity, digital infrastructure, fiscal incentives, and talent-attraction programs that lower the uncertainty and adjustment costs of transformation and raise the returns to expansion strategies that draw on young digital talent [31,43]. From a socio-economic systems perspective, policy pilots constitute the institutional layer of the social subsystem that conditions how the technical subsystem is deployed. We therefore expect supportive digital-economy policy pilots to strengthen the positive effect of intelligent transformation on youth employment.

H3c. *The positive effect of industrial intelligent transformation on youth employment is positively moderated by supportive digital-economy policy pilots.*

The conceptual framework of this study is illustrated in Figure 1.

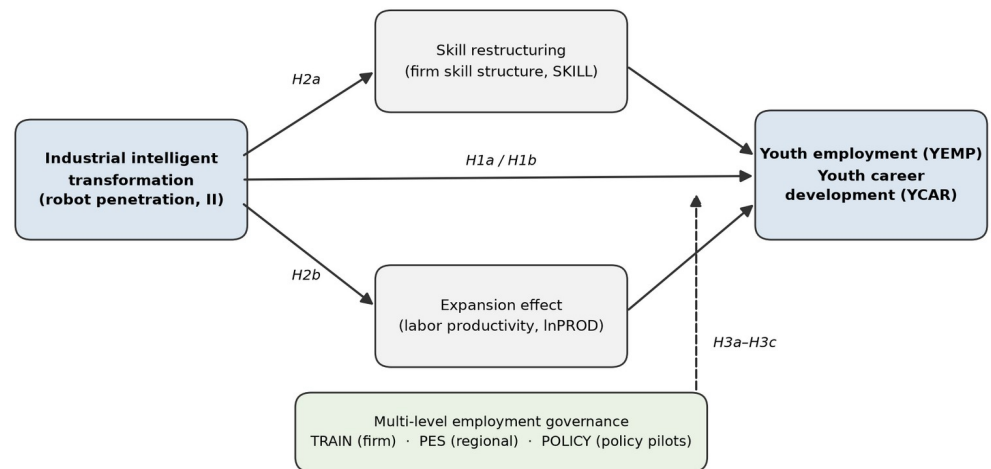


Figure 1. The conceptual framework.

3. Research Design

3.1. Variable Measurement

3.1.1. Youth Employment and Youth Career Development

Conceptually, youth employment refers to the firm-level demand for young workers, that is, the extent to which a firm employs workers in the early stage of their careers. Empirically, because employee age structure is disclosed with increasing coverage in the corporate-governance and employee files of Chinese listed firms, we measure youth employment (*YEMP*) as the share of a firm's employees aged 30 years and below in total employees, as defined in Equation (1):

$$YEMP_{i,t} = \frac{Emp_{30_{i,t}}}{Emp_{i,t}} \quad (1)$$

where $Emp_{30_{i,t}}$ is the number of employees aged 30 years and below in firm i in year t , and $Emp_{i,t}$ is the firm's total number of employees. Using a share rather than a level purges the measure of firm-size effects and captures the composition of labor demand toward young workers. Because a 35-year threshold is also commonly used in disclosure practice, we construct an alternative measure (*YEMP* 2), the share of employees aged 35 and below, for robustness. Employee age-structure and headcount data were obtained from the China Stock Market and Accounting Research Database (CSMAR) and supplemented by hand-collection from firms' annual reports.

Youth career development (*YCAR*) captures whether young talent advances into decision-making positions within the firm. Following the executive-profile literature, we measure *YCAR* as the proportion of a firm's directors, supervisors, and senior executives aged 40 years and below in the total executive team in year t , constructed from the CSMAR executive-characteristics files [28,37]. A higher value indicates that young professionals occupy a larger share of managerial positions, an observable marker of early-career advancement within the organizational hierarchy. As an alternative measure, we use the share of executives aged 45 and below.

3.1.2. Industrial Intelligent Transformation

Conceptually, industrial intelligent transformation refers to the firm-level transformation of production systems toward intelligent, adaptive, and data-enabled manufacturing. Empirically, because it is difficult to observe all dimensions of intelligent transformation at the firm-year level, we use industrial robot penetration as a proxy for the embodied, production-side dimension of intelligent transformation, following current research [14,34]. We construct a firm-level industrial intelligent transformation (II) measure based on robot penetration using a shift-share (Bartik) approach [44]. First, we construct the industry-level penetration rate of industrial robots in China according to Equation (2):

$$II_{m,t} = \frac{ROBOT_{m,t}}{L_{m,2010}} \quad (2)$$

where $II_{m,t}$ denotes the penetration rate of industrial robots in industry m in year t ; $ROBOT_{m,t}$ represents the operational stock of industrial robots in industry m in year t in China, drawn from the International Federation of Robotics (IFR) database; and $L_{m,2010}$ represents employment in the base year 2010 in industry m . Fixing the employment denominator at the pre-period level mitigates concerns about endogenous adjustments in industry employment. Second, we construct the firm-level exposure to robot penetration according to Equations (3) and (4):

$$\omega_{i,m} = \frac{RS_{i,m,2011}}{\overline{RS}_{m,2011}} \quad (3)$$

$$II_{i,t} = \omega_{i,m} \times II_{m,t} \quad (4)$$

where $II_{i,t}$ represents firm i 's exposure to industrial robot penetration in year t , and $\omega_{i,m}$ is a time-invariant exposure weight that captures firm i 's baseline intensity of production activities within industry m . $RS_{i,m,2011}$ denotes firm i 's share of employees in production-related positions in the base year 2011, and $\overline{RS}_{m,2011}$ denotes the industry- m median of the same measure in 2011. The ratio serves as the baseline exposure weight, which is interacted with the time-varying industry-level penetration $II_{m,t}$ to obtain the firm-year measure. As an alternative measure, we construct a text-based intelligent-transformation index (TII) from the frequency of intelligent-manufacturing-related keywords in firms' annual reports, following the text-measurement literature [37,45,46].

3.1.3. Mediating Variables

To unpack the channels through which intelligent transformation affects youth employment, we focus on two mediators: the firm skill structure and labor productivity.

Firm skill structure ($SKILL$). We proxy the firm skill structure by the share of employees holding a bachelor's degree or above in total employees, based on the education-structure data in CSMAR [15]. A higher value indicates a more skill-intensive workforce. As an alternative, we use the share of technical and R&D personnel in total employees.

Labor productivity ($lnPROD$). We measure the expansion channel by labor productivity, defined as the natural logarithm of operating revenue per

employee [13,40]. Productivity growth captures the cost-reduction and scale-expansion force through which automation enlarges the firm's activity and hiring capacity. As an alternative, we use total factor productivity estimated with the Levinsohn-Petrin method.

3.1.4. Moderating Variables

To test the moderating role of multi-level employment governance, we construct three variables spanning the firm, regional, and policy levels.

Firm training investment (*TRAIN*). We measure training investment as employee education and training expenses per employee, hand-collected from the notes to firms' financial statements, min-max normalized to the [0, 1] interval [35,41]. A higher value indicates a stronger internal skill-formation capability.

Regional public employment services (*PES*). We measure the intensity of public employment services in the prefecture-level city where the firm is registered by the natural logarithm of per-capita fiscal expenditure on education and employment-related programs, drawn from the China City Statistical Yearbook and provincial fiscal accounts [29,30]. This variable captures the local supply of job-placement platforms, vocational training, and employment subsidies available to young job seekers.

Digital-economy policy pilots (*POLICY*). We measure policy support using a dummy variable equal to 1 if the firm is registered in a city designated as a national big-data comprehensive pilot zone or a national new-generation artificial intelligence innovation and development pilot zone in year t , and 0 otherwise [25,31]. These national pilots provide regulatory support, digital infrastructure, and talent-attraction complementarities for intelligent transformation.

3.1.5. Control Variables

Following the extant literature, we control for a series of firm- and city-level variables that may affect youth employment. Firm-level controls include firm age (*Age*), firm size (*Size*), leverage (*Lev*), profitability (*RoA*), cash flow (*Cash*), sales growth (*Growth*), the shareholding of the top ten shareholders (*Top10*), board size (*Board*), CEO-chair separation (*Separation*), and state ownership (*SOE*). To account for regional economic conditions, we further include the natural logarithm of the gross domestic product of the city where the firm is registered (*lnGdp*), which controls for differences in local economic development, labor-market conditions, and demand that may be correlated with both intelligent transformation and youth employment. Detailed definitions of all variables are provided in Appendix A.1.

3.2. Regression Model Specification

3.2.1. Baseline Model

To examine the impact of industrial intelligent transformation on youth employment and career development, we estimate Equation (5):

$$Y_{i,t} = \alpha + \beta_1 II_{i,t-1} + \gamma' Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (5)$$

Here, i indexes firms and t indexes years. The dependent variable $Y_{i,t}$ alternatively denotes the youth-employment share $YEMP_{i,t}$ and the young-executive share $YCAR_{i,t}$ (Section 3.1.1). We use the one-year lag of

intelligent transformation, $II_{i,t-1}$, to mitigate simultaneity and reverse-causality concerns and to ensure that transformation is measured prior to labor outcomes (Section 3.1.2). $Controls_{i,t}$ includes the firm- and city-level controls. We include firm fixed effects μ_i and year fixed effects λ_t in a two-way fixed-effects design [47]; $\varepsilon_{i,t}$ is the error term, and standard errors are robust and clustered at the firm level. The coefficient of interest is β_1 ; H1a and H1b predict $\beta_1 > 0$.

3.2.2. Mediation Models

To test the mediating mechanisms, we estimate Equations (6) and (7):

$$M_{i,t} = \alpha + \delta_1 II_{i,t-1} + \theta' Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (6)$$

$$Y_{i,t} = \alpha + \varphi_1 M_{i,t} + \varphi_2 II_{i,t-1} + \kappa' Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (7)$$

$M_{i,t}$ denotes the mediator. For the skill-restructuring effect (H2a), we set $M_{i,t} = SKILL_{i,t}$; we expect $\delta_1 > 0$ and $\varphi_1 > 0$, indicating that intelligent transformation upgrades the firm skill structure and that a more skill-intensive structure raises youth employment. For the expansion effect (H2b), we set $M_{i,t} = lnPROD_{i,t}$; we expect $\delta_1 > 0$ and $\varphi_1 > 0$, indicating that transformation raises labor productivity and that productivity-driven expansion promotes youth employment. We report bootstrap confidence intervals for the indirect effects to assess the statistical significance of mediation [48,49].

3.2.3. Moderation Models

To examine the moderating role of multi-level employment governance, we estimate Equation (8):

$$Y_{i,t} = \alpha + \eta_1 II_{i,t-1} + \eta_2 Mod_{i,t} + \eta_3 II_{i,t-1} \times Mod_{i,t} + \vartheta' Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (8)$$

$Mod_{i,t}$ alternatively captures firm training investment (*TRAIN*), regional public employment services (*PES*), and digital-economy policy pilots (*POLICY*). The coefficient of interest is η_3 , which corresponds to H3a–H3c and is expected to be positive, indicating that stronger employment governance amplifies the positive effect of intelligent transformation on youth employment. Continuous moderators are mean-centered before constructing the interaction terms.

3.3. Data and Sample Selection

We construct a panel of A-share listed firms registered in Zhejiang province from 2011 to 2023. Zhejiang hosts one of the largest provincial populations of listed firms in China, dominated by private manufacturing enterprises, which makes the sample well suited to studying the labor-market consequences of regional industrial upgrading. Following standard practice, we exclude: (1) specially treated firms (ST, *ST, or PT); (2) firms in the financial industry; and (3) firm-year observations with missing key variables. Firm financial, employee, executive, and governance data were obtained from CSMAR; industry robot data were obtained from the IFR database; training expenses were hand-collected from annual reports; and city-level fiscal and GDP data were obtained from the China City Statistical Yearbook. All continuous variables were winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. Because the IFR robot stock is available from 2010 onward, the one-year-lagged transformation measure is defined

for every sample year. The final sample comprises 6,124 firm-year observations for 668 unique firms.

3.4. Descriptive Statistics

Table 1 reports the descriptive statistics for all variables. The mean youth-employment share is 0.312, indicating that employees aged 30 and below account for roughly one-third of the workforce of Zhejiang listed firms, with substantial cross-firm variation (standard deviation 0.147). The mean young-executive share is 0.186. Table 2 presents the correlation matrix; the correlations suggest a positive association between intelligent transformation and both youth-employment outcomes. Appendix A.2 reports the variance inflation factor (VIF) values, with a mean VIF of 1.53, indicating that multicollinearity is not a serious concern.

Table 1. Descriptive statistics for variables.

Variables	Obs	Mean	Std. Dev.	Min	Max
YEMP	6,124	0.312	0.147	0	0.826
YCAR	6,124	0.186	0.118	0	0.667
II	6,124	0.573	0.804	0	4.116
SKILL	6,124	0.247	0.169	0.021	0.913
lnPROD	6,124	13.62	0.94	11.08	16.47
TRAIN	6,124	0.318	0.201	0	1
PES	6,124	8.74	0.52	7.36	10.01
POLICY	6,124	0.462	0.499	0	1
Age	6,124	2.87	0.34	1.61	3.53
Size	6,124	22.14	1.18	19.87	26.32
Lev	6,124	0.417	0.196	0.048	0.892
RoA	6,124	0.043	0.062	−0.284	0.226
Cash	6,124	0.048	0.071	−0.163	0.251
Growth	6,124	0.146	0.352	−0.517	2.418
Top10	6,124	0.588	0.147	0.216	0.924
Board	6,124	2.11	0.19	1.61	2.71
Separation	6,124	0.281	0.450	0	1
SOE	6,124	0.194	0.396	0	1
lngdp	6,124	8.92	0.63	7.44	9.86

Notes: YEMP is the youth-employment share; YCAR is the young-executive share; II is industrial intelligent transformation. All variable definitions are provided in Appendix A.1.

Table 2. Correlation matrix.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1) YEMP	1.000							
(2) YCAR	0.214 ***	1.000						
(3) II	0.118 ***	0.087 ***	1.000					
(4) SKILL	0.126 ***	0.153 ***	0.204 ***	1.000				
(5) lnPROD	0.095 ***	0.061 ***	0.172 ***	0.238 ***	1.000			
(6) TRAIN	0.083 ***	0.079 ***	0.146 ***	0.187 ***	0.117 ***	1.000		
(7) PES	0.071 ***	0.048 ***	0.139 ***	0.162 ***	0.128 ***	0.094 ***	1.000	
(8) POLICY	0.064 ***	0.042 ***	0.151 ***	0.148 ***	0.106 ***	0.088 ***	0.312 ***	1.000

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

4. Empirical Results

4.1. Baseline Regression Results

We estimate Equation (5) to examine the effect of industrial intelligent transformation on youth employment and career development; Table 3 reports the baseline results. Columns (1) and (2) take the youth-employment share as the dependent variable, presenting the parsimonious specification with firm and year fixed effects and the full specification with firm- and city-level controls, respectively. Columns (3) and (4) repeat this structure for the young-executive share. Columns (5) and (6) replace year fixed effects with

industry-by-year fixed effects to control for time-varying industry shocks. Across all specifications, the coefficient on lagged intelligent transformation is positive and statistically significant. For youth employment, the estimated coefficient ranges from 0.0112 to 0.0127 and remains significant at the 1% level after adding controls and more stringent fixed effects, supporting H1a. For the young-executive share, the coefficient ranges from 0.0068 to 0.0078 and remains significant, supporting H1b.

Beyond statistical significance, the results are economically meaningful. Taking the full specification in Column (2) as an example, a one-standard-deviation increase in intelligent transformation is associated with an increase of approximately 0.96 percentage points in the youth-employment share, equivalent to about 3.1% of the sample mean. The corresponding effect on the young-executive share (Column (4)) is 0.57 percentage points, also about 3.1% of its mean. From a socio-economic systems perspective, these results suggest that intelligent transformation acts as a technological subsystem shock that expands both the demand for young, digitally skilled workers and the advancement opportunities of young managerial talent, rather than simply displacing labor [13,20].

Table 3. Baseline regression results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	YEMP	YEMP	YCAR	YCAR	YEMP	YCAR
L.II	0.0127 *** (0.0035)	0.0119 *** (0.0036)	0.0078 *** (0.0027)	0.0071 ** (0.0028)	0.0112 *** (0.0037)	0.0068 ** (0.0029)
Controls	No	Yes	No	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	No	No
Industry × Year FE	No	No	No	No	Yes	Yes
Observations	6,124	6,124	6,124	6,124	6,124	6,124
R-squared	0.612	0.634	0.548	0.567	0.651	0.583

Notes: All columns use one-year-lagged II (L.II). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Robust standard errors clustered at the firm level are reported in parentheses.

From a practical perspective, industrial intelligent transformation does not aggravate the youth employment dilemma at the firm level. Instead, as Zhejiang firms embed robots and intelligent equipment into their production processes, they create new digitally intensive roles, recruit young technical talent, and promote young managers, indicating that the regional dilemma stems less from technology itself than from frictions in skill formation and matching—an interpretation we examine directly in the mechanism and moderation analyses below.

4.2. Robustness Checks

4.2.1. Alternative Measures, Sample Restrictions, and Fixed Effects

To further assess robustness, we re-estimate Equation (5) using alternative measures and subsamples; Table 4 summarizes the results for the youth-employment share. First, Column (1) replaces the dependent variable with the alternative measure based on the 35-year threshold (*YEMP2*); the coefficient on intelligent transformation remains positive and significant. Second, Column (2) replaces the explanatory variable with the text-based intelligent-transformation index (*TII*), and the result is unchanged. Third, Column (3) excludes the year 2020 to rule out potential confounding effects of the COVID-19 shock, and the estimated effect remains significantly positive. Fourth, Column (4) excludes firms listed after 2018 to mitigate

composition effects from new listings, and the result is robust. Fifth, Column (5) restricts the sample to a balanced panel of firms observed in every sample year, mitigating concerns about entry and exit, and the coefficient remains positive and significant. Sixth, Column (6) reports two-way clustered standard errors at the firm and city levels. Finally, Column (7) further includes city-by-year fixed effects to control for time-varying local shocks, such as city-level talent-attraction policies. Across all specifications, the estimated effect of intelligent transformation on youth employment is consistently positive and statistically significant.

Table 4. Robustness checks: alternative measures, sample restrictions, and fixed effects.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Alt. DV (YEMP2)	Alt. IV (TII)	Excl. 2020	Excl. new listings	Balanced panel	Two-way cluster	City × Year FE
L.II	0.0148 *** (0.0044)	0.0086 *** (0.0027)	0.0121 *** (0.0038)	0.0115 *** (0.0039)	0.0126 *** (0.0043)	0.0119 ** (0.0047)	0.0108 *** (0.0040)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Observations	5,893	6,124	5,652	5,417	4,551	6,124	6,124
R-squared	0.618	0.629	0.641	0.639	0.643	0.634	0.662

Notes: Column (2) uses the text-based intelligent-transformation index as the explanatory variable. Column (6) reports two-way clustered standard errors (firm and city). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Robust standard errors clustered at the firm level are reported in parentheses unless otherwise noted.

4.2.2. Propensity Score Matching

Our baseline estimates may be subject to endogeneity from firms' self-selection into intelligent transformation. To address this concern, we employ propensity score matching (PSM). We define II_{dummy} as an indicator equal to 1 for firms with intelligent transformation above the annual median and 0 otherwise, and estimate a probit model in the first stage using the baseline covariates and industry fixed effects. We implement one-to-one nearest-neighbor matching without replacement and impose a caliper of 0.01 to enhance comparability between the treatment and control groups [50]. Table 5 reports the results before and after matching. The coefficient on II_{dummy} remains positive and statistically significant after matching, suggesting that our main finding is not driven by observable differences between high- and low-transformation firms. Figure 2 further shows improved covariate balance after matching.

Table 5. Robustness check: propensity score matching.

Variables	(1)	(2)
	Pre-Match YEMP	Post-Match YEMP
II_dummy	0.0102 * (0.0058)	0.0131 ** (0.0061)
Age	−0.0241 *** (0.0056)	−0.0228 *** (0.0060)
Size	−0.0119 *** (0.0031)	−0.0113 *** (0.0034)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	6,124	4,682
R-squared	0.634	0.629

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Robust standard errors clustered at the firm level are reported in parentheses.

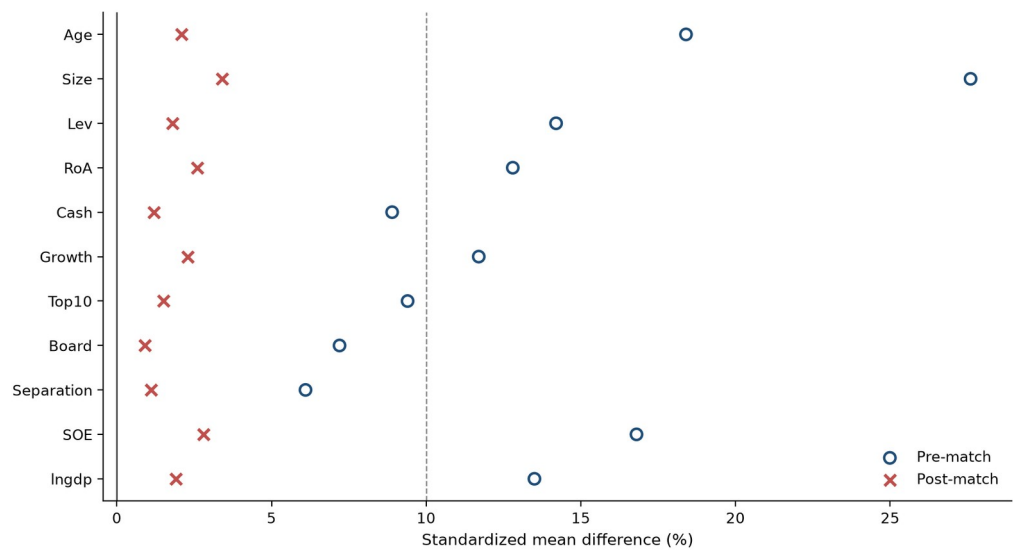


Figure 2. Standardized mean differences of covariates: pre-match vs. post-match.

4.2.3. Instrumental Variable Regression

To further alleviate endogeneity concerns, we employ a two-stage least squares (2SLS) instrumental-variable approach. Following the shift-share (Bartik) tradition [44], we construct the first instrument (IV1) as the robot penetration of the same industry in other East Asian economies interacted with the firm's baseline production exposure. Foreign industry-level robot adoption reflects global technological trends that affect the focal firm's transformation but are unlikely to directly determine its youth-employment share [9,13]. Following the literature that uses geographic and infrastructural conditions as exogenous drivers of technology adoption [25,51], we use the second instrument (IV2), the historical density of fixed telephone lines in the firm's city interacted with a time trend, capturing local digital-infrastructure endowments. Table 6 reports the results. In the first stage, both instruments are positive and statistically significant, with Kleibergen–Paap first-stage F-statistics well above the weak-instrument threshold. In the second stage, the fitted value of intelligent transformation remains positively and significantly associated with youth employment, consistent with the baseline findings.

Table 6. Robustness check: instrumental-variable regression.

Variables	(1)	(2)	(3)	(4)
	First Stage	Second Stage	First Stage	Second Stage
	L.II	YEMP	L.II	YEMP
IV1 (foreign robots)	0.386 *** (0.052)			
IV2 (infrastructure)			0.242 *** (0.043)	
L.II (fitted)		0.0217 *** (0.0071)		0.0195 ** (0.0084)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
K-P F-statistic	54.7		31.6	
Observations	6,124	6,124	6,124	6,124

Notes: Kleibergen–Paap first-stage F-statistics exceed the Stock–Yogo weak-instrument critical values. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Robust standard errors clustered at the firm level are reported in parentheses.

4.3. Mechanism Analysis

Having established that intelligent transformation promotes youth employment and career development, we examine the underlying mechanisms. Drawing on our hypotheses, we consider two mediating channels: the firm skill structure (skill-restructuring effect) and labor productivity (expansion effect). We estimate the mediation models in Equations (6) and (7); Table 7 reports the results.

Columns (1) and (2) report the skill-restructuring channel. Column (1) shows that intelligent transformation significantly upgrades the firm skill structure ($\delta_1 = 0.0436$, $p < 0.01$). Column (2) shows that a more skill-intensive structure is positively and significantly associated with youth employment ($\varphi_1 = 0.048$, $p < 0.01$), while the coefficient on intelligent transformation remains positive, consistent with partial mediation. These results support H2a, indicating that intelligent transformation promotes youth employment partly by raising the demand for young, high-skill workers. Columns (3) and (4) report the expansion channel. Column (3) shows that intelligent transformation significantly raises labor productivity ($\delta_1 = 0.0571$, $p < 0.01$), and Column (4) shows that higher productivity is positively and significantly associated with youth employment ($\varphi_1 = 0.0089$, $p < 0.01$). This supports H2b. Bootstrap tests confirm that both indirect effects are statistically significant. A comparison of the indirect effects indicates that the skill-restructuring channel accounts for a larger share of the mediated effect than the expansion channel, suggesting that intelligent transformation promotes youth employment mainly by reconfiguring the skill composition of labor demand, with productivity-driven expansion serving as a complementary channel.

Table 7. Mechanism analysis results.

Variables	(1)	(2)	(3)	(4)
	SKILL	YEMP	lnPROD	YEMP
L.II	0.0436 *** (0.0072)	0.0098 *** (0.0036)	0.0571 *** (0.0148)	0.0106 *** (0.0037)
SKILL		0.048 *** (0.011)		
lnPROD				0.0089 *** (0.0021)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	6,124	6,124	6,124	6,124
R-squared	0.703	0.641	0.688	0.638

Notes: Columns (1)–(2) report the skill-restructuring channel; Columns (3)–(4) report the expansion (productivity) channel. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Robust standard errors clustered at the firm level are reported in parentheses.

4.4. Moderating Effects of Multi-Level Employment Governance

Table 8 reports the moderating role of multi-level employment governance. Consistent with H3a–H3c, the interaction terms between intelligent transformation and the governance variables are positive and statistically significant. Column (1) shows that the interaction $L.II \times TRAIN$ is positive and significant ($\eta_3 = 0.0311$, $p < 0.05$), indicating that firms with stronger training investment convert transformation into youth hiring more effectively, supporting H3a. Column (2) shows that the interaction $L.II \times PES$ is positive and significant ($\eta_3 = 0.0158$, $p < 0.05$), supporting

H3b and indicating that regional public employment services reinforce the youth-employment benefits of transformation. Column (3) shows that the interaction $L.II \times POLICY$ is positive and significant ($\eta_3 = 0.0136$, $p < 0.05$), supporting H3c and indicating that digital-economy policy pilots strengthen the transformation-youth-employment linkage. From a socio-economic systems perspective, these results show that employment governance is a key social-subsystem condition that helps regional systems translate industrial upgrading into youth-inclusive opportunity, and that differences in governance capacity partly explain heterogeneous youth-employment responses across firms and cities [29,31].

Table 8. The moderating role of multi-level employment governance.

Variables	(1)	(2)	(3)
	YEMP	YEMP	YEMP
L.II	0.0104 *** (0.0038)	0.0109 *** (0.0037)	0.0113 *** (0.0036)
TRAIN	0.0218 ** (0.0104)		
L.II $\times$ TRAIN	0.0311 ** (0.0128)		
PES		0.0125 (0.0093)	
L.II $\times$ PES		0.0158 ** (0.0071)	
POLICY			0.0158 ** (0.0067)
L.II $\times$ POLICY			0.0136 ** (0.0062)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	6,124	6,124	6,124
R-squared	0.638	0.636	0.637

Notes: TRAIN = firm training investment; PES = regional public employment services; POLICY = digital-economy policy pilots. Continuous moderators are mean-centered. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Robust standard errors clustered at the firm level are reported in parentheses.

4.5. Heterogeneity Analysis

Building on the benchmark results, we explore whether the effect of intelligent transformation on youth employment differs across firms and locations within the regional system. We conduct subsample regressions along four dimensions: industry, firm size, ownership, and intra-provincial location. The results are reported in Table 9. Columns (1) and (2) show that the positive effect is significant for manufacturing firms but insignificant for services firms, consistent with the production-side nature of robot-based transformation. Columns (3) and (4) show that the effect is significant for large firms but not for small firms, reflecting differences in transformation capacity and internal labor markets. Columns (5) and (6) show that the effect is significant for non-state-owned firms but not for state-owned firms, consistent with stronger market-driven hiring flexibility in Zhejiang's private sector. Columns (7) and (8) show that the effect is stronger for firms registered in the digital core cities (Hangzhou and Ningbo) than for firms in the other cities of the province.

Overall, the heterogeneity results suggest that the youth-employment benefits of intelligent transformation are concentrated among manufacturing, larger, non-state-owned, and digital-core-city firms. From a

socio-economic systems perspective, these findings indicate that firms' youth-employment responses depend on the alignment among technological capacity, resource endowments, market-driven governance, and supportive regional ecosystems. Manufacturing and large firms possess greater capacity to reconfigure production systems and job structures; non-state-owned firms have stronger incentives and flexibility in recruitment; and digital-core-city firms benefit from denser digital infrastructure, talent pools, and policy experimentation [11,31]. Industrial intelligent transformation is therefore more likely to be translated into youth employment when firms have both stronger internal capabilities and stronger external ecosystem support for socio-technical adaptation.

Table 9. Heterogeneity analysis results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Manuf.	Services	Large	Small	Non-SOE	SOE	Core cities	Other cities
L.II	0.0141 *** (0.0042)	0.0054 (0.0058)	0.0138 *** (0.0043)	0.0052 (0.0064)	0.0132 *** (0.0040)	0.0061 (0.0069)	0.0149 *** (0.0046)	0.0071 * (0.0042)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,238	1,886	3,062	3,062	4,936	1,188	2,975	3,149
R-squared	0.647	0.619	0.652	0.611	0.641	0.628	0.649	0.622

Notes: Core cities are Hangzhou and Ningbo; large (small) firms are those above (below) the annual median of total assets. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Robust standard errors clustered at the firm level are reported in parentheses.

5. Discussion and Limitations

5.1. Discussion

First, this study shows that industrial intelligent transformation significantly promotes both youth employment and youth career development in Zhejiang listed firms. This finding suggests that, at the firm level, the creation and reinstatement forces of intelligent transformation outweigh its displacement force for young workers [8,13]. Unlike the pessimistic narrative that frames automation as a wholesale threat to young people's jobs, our evidence indicates that the youth employment dilemma of a rapidly upgrading region does not originate from firm-level technology adoption itself. Rather, youth employment can be understood as an adaptive outcome through which firms reconfigure tasks, skills, and authority structures under technological change. The positive effect on the young-executive share further indicates that technological renewal opens advancement opportunities for young managerial and technical talent, linking the employment and career-development dimensions of the regional dilemma [16,28].

Second, the mechanism analysis shows that this effect operates through two resource-reconfiguration pathways. The skill-restructuring effect indicates that intelligent transformation upgrades the firm skill structure, raising the relative demand for young, high-skill workers in which recent graduates hold a comparative advantage [6,15]. The expansion effect shows that transformation raises labor productivity and enlarges the firm's scale of activity, expanding total hiring capacity that disproportionately absorbs young job seekers [14,40]. These findings locate the root of the youth employment dilemma in the coupling between skill formation and task creation: when the regional skill-formation subsystem keeps pace with the

transformed technical subsystem, industrial upgrading absorbs rather than excludes young workers.

Third, the moderating results highlight the role of multi-level employment governance as a social-subsystem condition. Firms with stronger training investment, cities with better public employment services, and locations covered by digital-economy policy pilots translate intelligent transformation into youth employment more effectively [29,31,35]. This finding supports the view that technological change alone does not determine labor outcomes; rather, firm, regional, and policy governance jointly shape whether industrial upgrading produces youth-inclusive value creation. It also provides direct empirical support for the countermeasure logic embedded in regional socio-economic system analysis: interventions at multiple system levels are complementary rather than substitutable [21,33].

Finally, the heterogeneity analysis identifies the boundary conditions of this relationship. The effect of intelligent transformation is stronger among manufacturing, larger, non-state-owned, and digital-core-city firms, which possess stronger transformation capabilities, recruitment flexibility, or ecosystem support. These findings further suggest that youth employment emerges from the interaction among technological change, skill reconfiguration, organizational renewal, and multi-level governance, reflecting a dynamic socio-economic adaptation process within the regional system [17,27].

5.2. Limitations and Future Research

This study has several limitations that provide directions for future research. First, although we measure youth employment using the share of employees aged 30 and below and adopt an alternative age threshold for robustness, this measure captures the age composition of the workforce rather than gross youth hiring and separation flows. Similarly, the young-executive share captures the stock of young managers rather than individual promotion events. Future research may use linked employer-employee or social-security data to examine youth hiring, retention, promotion, and job quality more directly.

Second, our intelligent-transformation measure is a robot-based exposure proxy that captures the embodied, production-side dimension of transformation, and the text-based alternative captures disclosure emphasis in addition to realized activity. Future research may combine deployment-level robot data, detailed technology investment records, and micro-level task data to better capture the multidimensional nature of industrial intelligence [37,46].

Third, this study focuses on listed firms registered in a single, economically advanced province. Listed firms are larger and more formalized than the small and medium-sized enterprises that employ most young workers, and Zhejiang's governance environment may not be representative of less developed regions. The external validity of the findings should therefore be interpreted with caution. Future research may extend the analysis to unlisted small and medium-sized enterprises, compare provinces at different stages of industrial upgrading, and examine whether intelligent

transformation improves the wages, job quality, and long-term career trajectories of young workers [5,52].

6. Conclusions and Implications

6.1. Conclusions

Using panel data on 668 A-share listed firms registered in Zhejiang province from 2011 to 2023, this study examined the relationship between industrial intelligent transformation and youth employment and career development from a regional socio-economic systems perspective, with particular attention to skill restructuring, productivity-driven expansion, and multi-level employment governance. The results show that intelligent transformation significantly increases both the youth-employment share and the young-executive share, and these findings remain robust across a range of tests, including propensity score matching and instrumental-variable estimation.

The mechanism analysis shows that intelligent transformation promotes youth employment through two pathways: a skill-restructuring effect through the upgrading of the firm skill structure and an expansion effect through productivity-driven growth. The moderation analysis further shows that multi-level employment governance strengthens this relationship: the positive effect of intelligent transformation is stronger in firms with higher training investment, in cities with stronger public employment services, and in locations covered by digital-economy policy pilots. The heterogeneity analysis shows that the effect is more pronounced among manufacturing, larger, non-state-owned, and digital-core-city firms. These findings indicate that the youth employment dilemma is not a mechanical by-product of industrial upgrading, but an emergent outcome of a regional socio-economic system shaped by technological change, skill reconfiguration, organizational renewal, and multi-level governance.

6.2. Implications

This study provides several practical implications for resolving the youth employment dilemma in regions undergoing rapid industrial upgrading. First, firms should treat intelligent transformation as an opportunity for youth-inclusive organizational renewal rather than solely as a tool for labor cost reduction. By coupling robot adoption with the creation of new technical roles, campus recruitment, and fast-track promotion channels for young talent, firms can align technological upgrading with the expansion of youth employment and career development.

Second, firms should strengthen internal skill-formation systems. Increasing training investment, building structured apprenticeship and mentoring programs around intelligent production lines, and co-designing curricula with vocational colleges can lower the skill-mismatch frictions that prevent young job seekers from filling newly created positions. This is especially important for small and traditional firms, in which the youth-employment benefits of transformation are currently weakest.

Third, local governments should upgrade regional public employment services into a system-level matching infrastructure. Integrating public job-placement platforms with industry skill demand forecasts, expanding

subsidized vocational training aligned with intelligent manufacturing, and providing employment subsidies targeted at young workers can strengthen the coupling between the regional skill-formation subsystem and upgrading firms. Provincial coordination can extend the governance capacity of the digital core cities to the other cities of the province, narrowing intra-regional disparities in youth opportunity.

Finally, policymakers should exploit digital-economy policy pilots as vehicles for youth-inclusive industrial upgrading. Embedding youth-employment objectives, talent-attraction programs, and training complementarities into big-data and artificial intelligence pilot zones can ensure that institutional support for intelligent transformation simultaneously supports the young labor force. Coordinating industrial, digital-economy, education, and labor-market policies within the regional system can turn the youth employment dilemma into a virtuous cycle of technological upgrading, skill formation, and inclusive growth.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Appendix A.1

Table A1 presents the detailed definitions of all variables.

Table A1. Variable definitions.

Variable	Definition
YEMP	Youth employment. Share of employees aged 30 years and below in total employees.
YEMP2	Alternative youth employment. Share of employees aged 35 years and below in total employees.
YCAR	Youth career development. Share of directors, supervisors, and senior executives aged 40 and below in the executive team.
II	Industrial intelligent transformation. Firm-level exposure to industrial robot penetration constructed with the shift-share approach in Equations (2)–(4).
TII	Alternative transformation measure. Natural logarithm of one plus the frequency of intelligent-manufacturing keywords in the annual report.
SKILL	Firm skill structure. Share of employees holding a bachelor's degree or above in total employees.
lnPROD	Labor productivity. Natural logarithm of operating revenue per employee.
TRAIN	Firm training investment. Employee education and training expenses per employee, min-max normalized to [0, 1].

Variable	Definition
PES	Regional public employment services. Natural logarithm of city per-capita fiscal expenditure on education and employment programs.
POLICY	Digital-economy policy pilots. Dummy equal to 1 if the firm’s city is a national big-data or AI innovation pilot zone in year t.
Age	Firm age. Natural logarithm of one plus the number of years since establishment.
Size	Firm size. Natural logarithm of total assets.
Lev	Leverage. Total liabilities divided by total assets.
RoA	Profitability. Net profit divided by total assets.
Cash	Cash flow. Net operating cash flow divided by total assets.
Growth	Sales growth. Annual growth rate of operating revenue.
Top10	Ownership concentration. Shareholding ratio of the top ten shareholders.
Board	Board size. Natural logarithm of the number of board members.
Separation	CEO-chair separation. Dummy equal to 1 if the CEO and board chair are different persons.
SOE	State ownership. Dummy equal to 1 for state-owned enterprises.
lngdp	City economic development. Natural logarithm of the GDP of the city where the firm is registered.

Appendix A.2

Table A2 reports the variance inflation factor (VIF) test for the main explanatory and control variables. The maximum VIF is 2.34 and the mean VIF is 1.53, both well below the commonly used threshold of 10, suggesting that multicollinearity is not a serious concern.

Table A2. Variance inflation factor test.

Variables	VIF	1/VIF
L.II	1.24	0.806
Age	1.38	0.725
Size	2.34	0.427
Lev	1.86	0.538
RoA	1.52	0.658
Cash	1.31	0.763
Growth	1.19	0.840
Top10	1.42	0.704
Board	1.27	0.787
Separation	1.12	0.893
SOE	1.61	0.621
lngdp	2.07	0.483
Mean VIF	1.53	

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